

Basics of Material Drying Technology

Hosokawa Educational Center



HOSOKAWA MICRON POWDER SYSTEMS

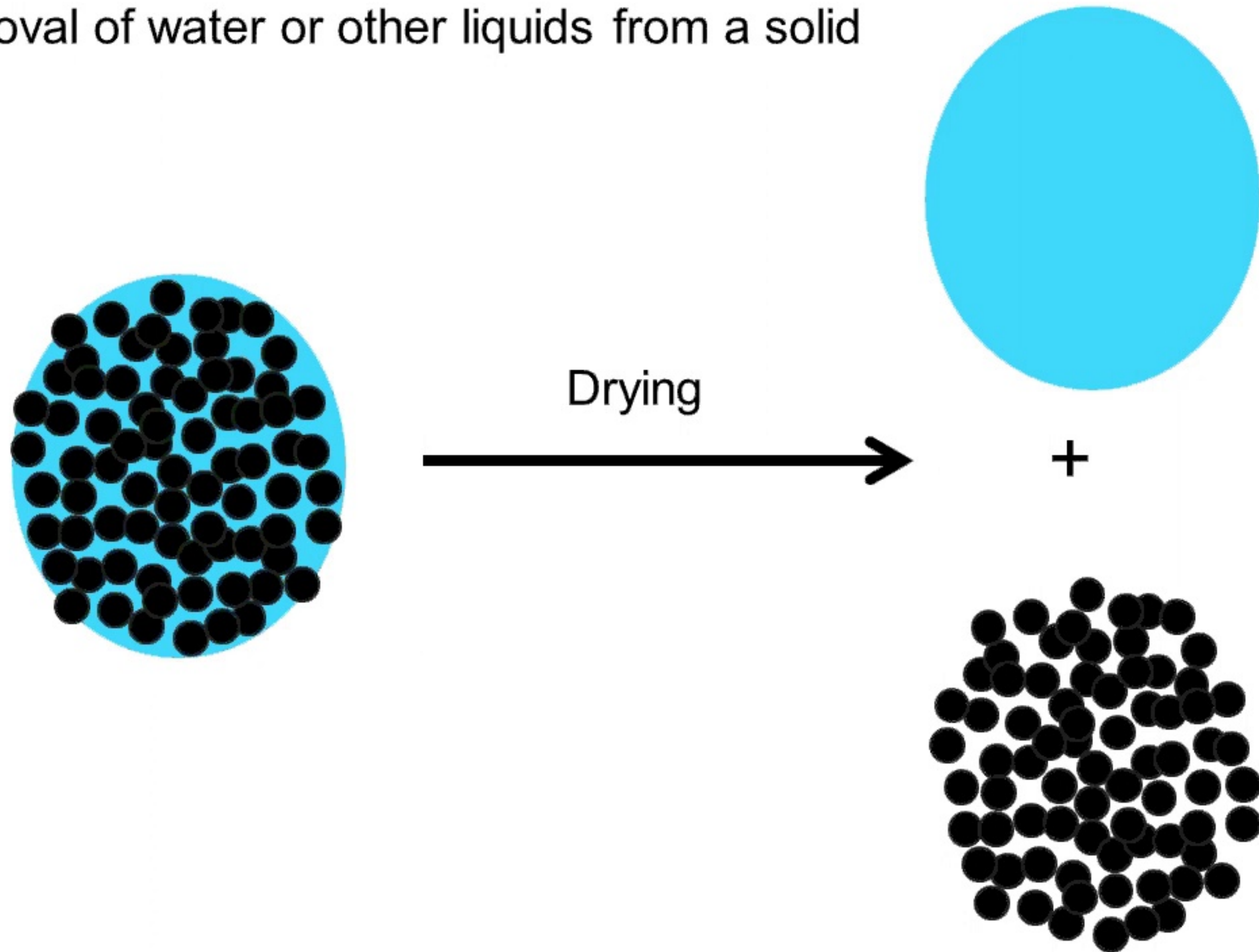
Table of Contents

- The basics of drying principles
- Theories for flash, vacuum and freeze drying
- Various types of dryers Hosokawa offers
- Review drying systems for a variety of material applications



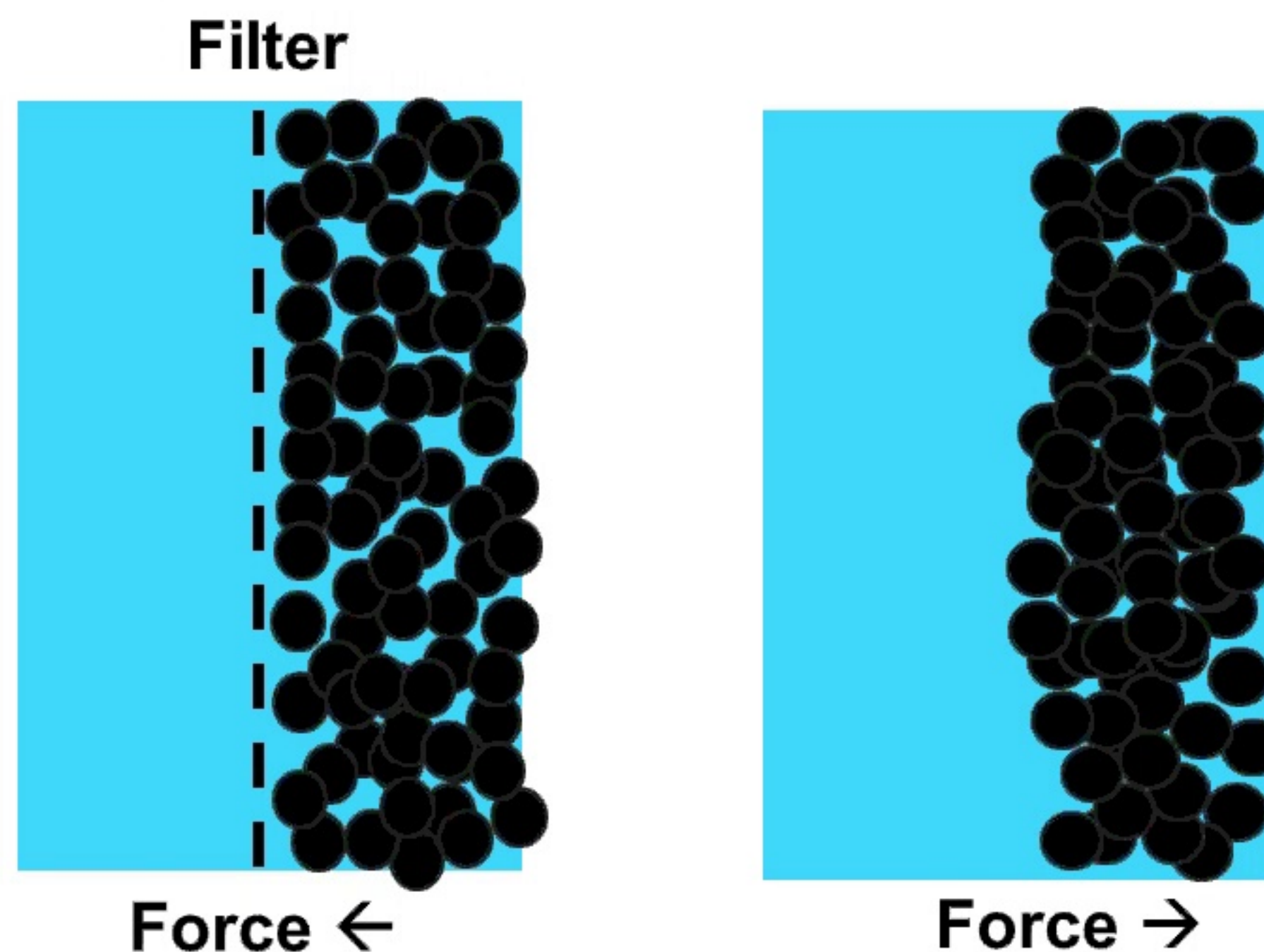
What is Drying?

- The removal of water or other liquids from a solid



How to remove the liquid?

- Removal by mechanical means:
 - **Filtration**
 - **Centrifugation**



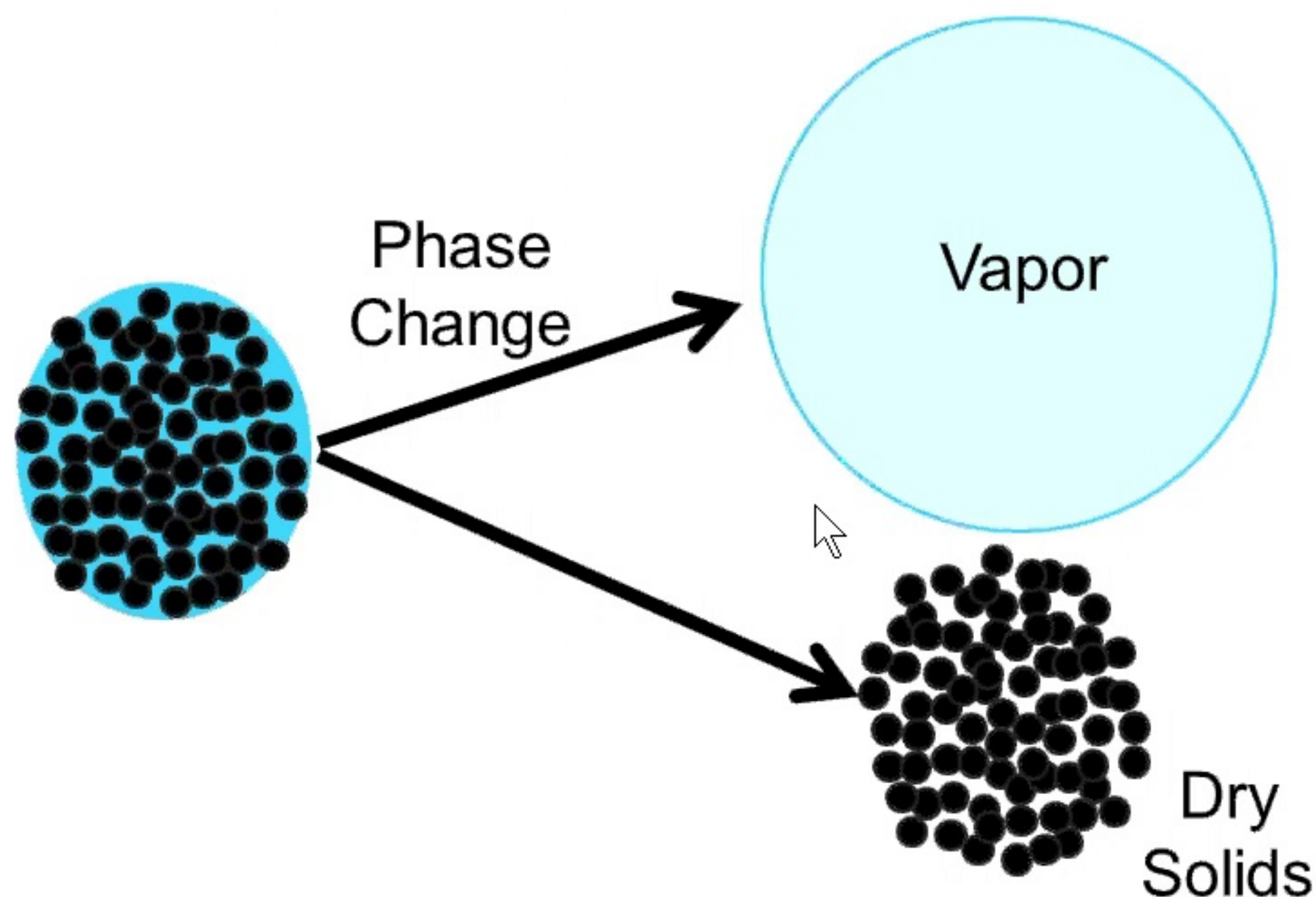
Mechanical Drying

- **Uses Mechanical Force to separate the solids and liquids**
- **Advantages**
 - Low energy usage
- **Disadvantages**
 - Limits on final moisture content



How to remove the liquid?

- Removal by thermodynamic means
 - Evaporation
 - Flash and Vacuum Drying
 - Sublimation (Lyophilization)
 - Freeze Drying

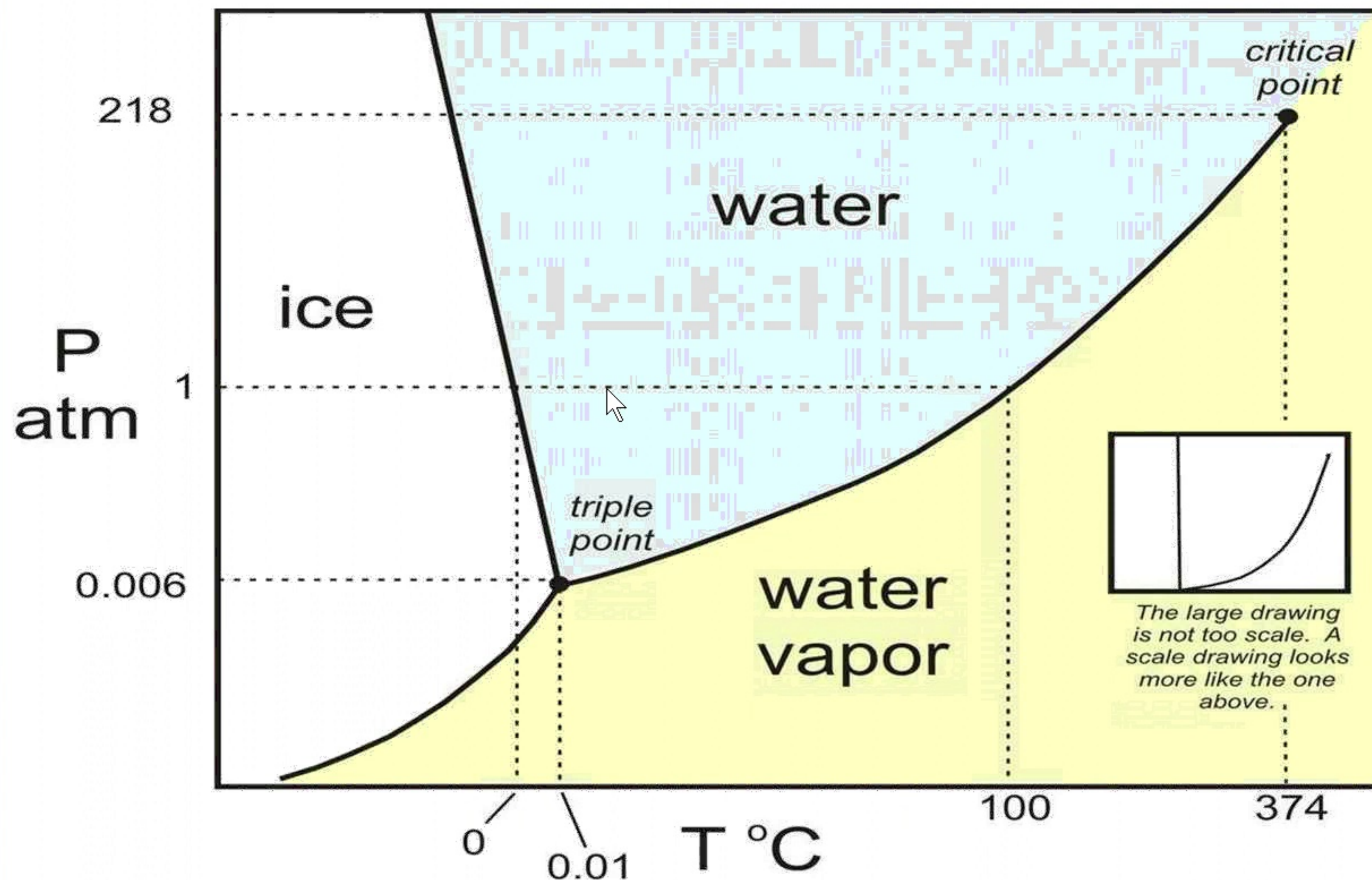


Thermodynamic Drying

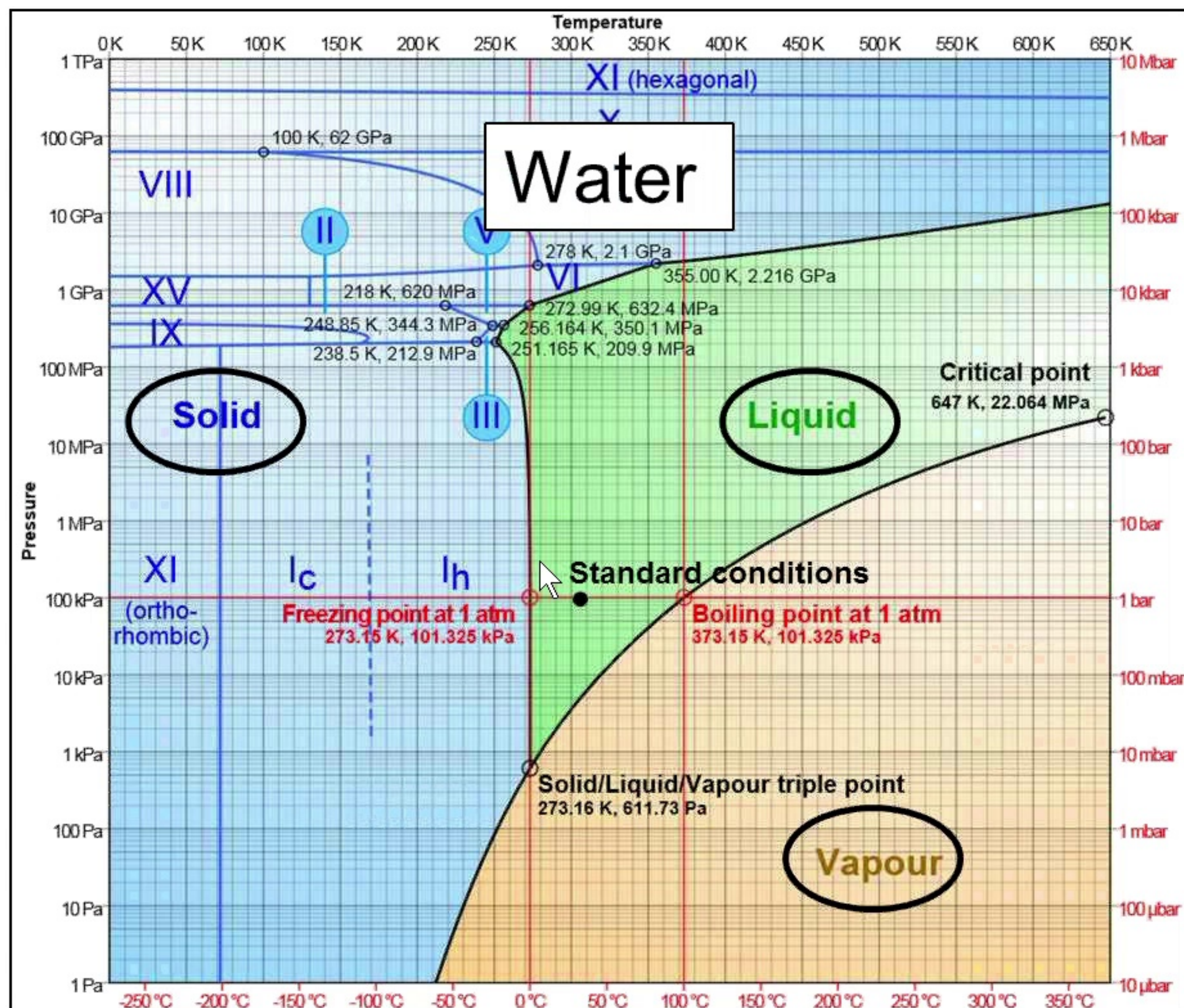
- Changing the liquid to a vapor, possibly after changing to a solid first
- Advantages
 - Low final moisture content possible
- Disadvantages
 - Higher energy usage



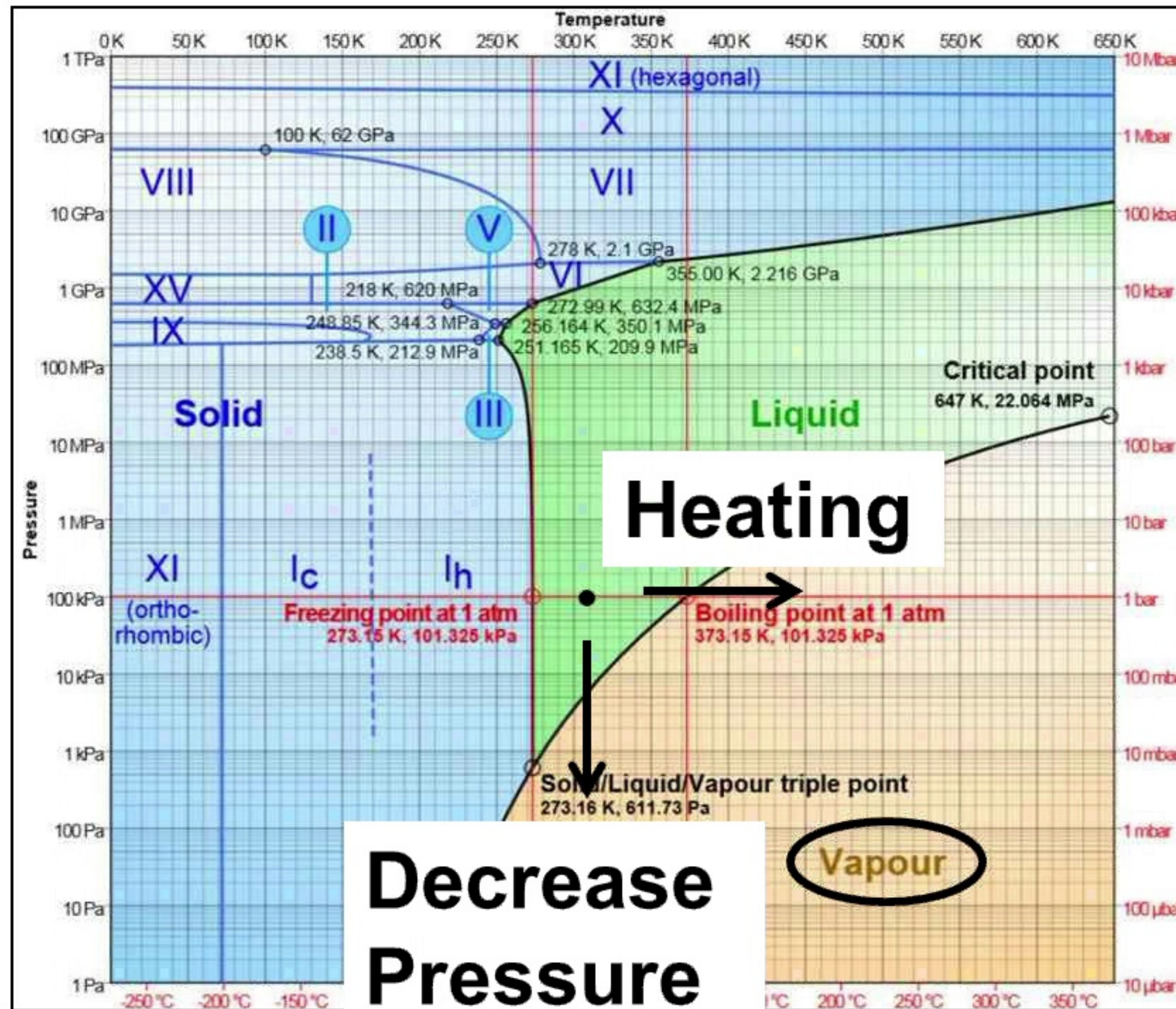
Water Phase Diagram



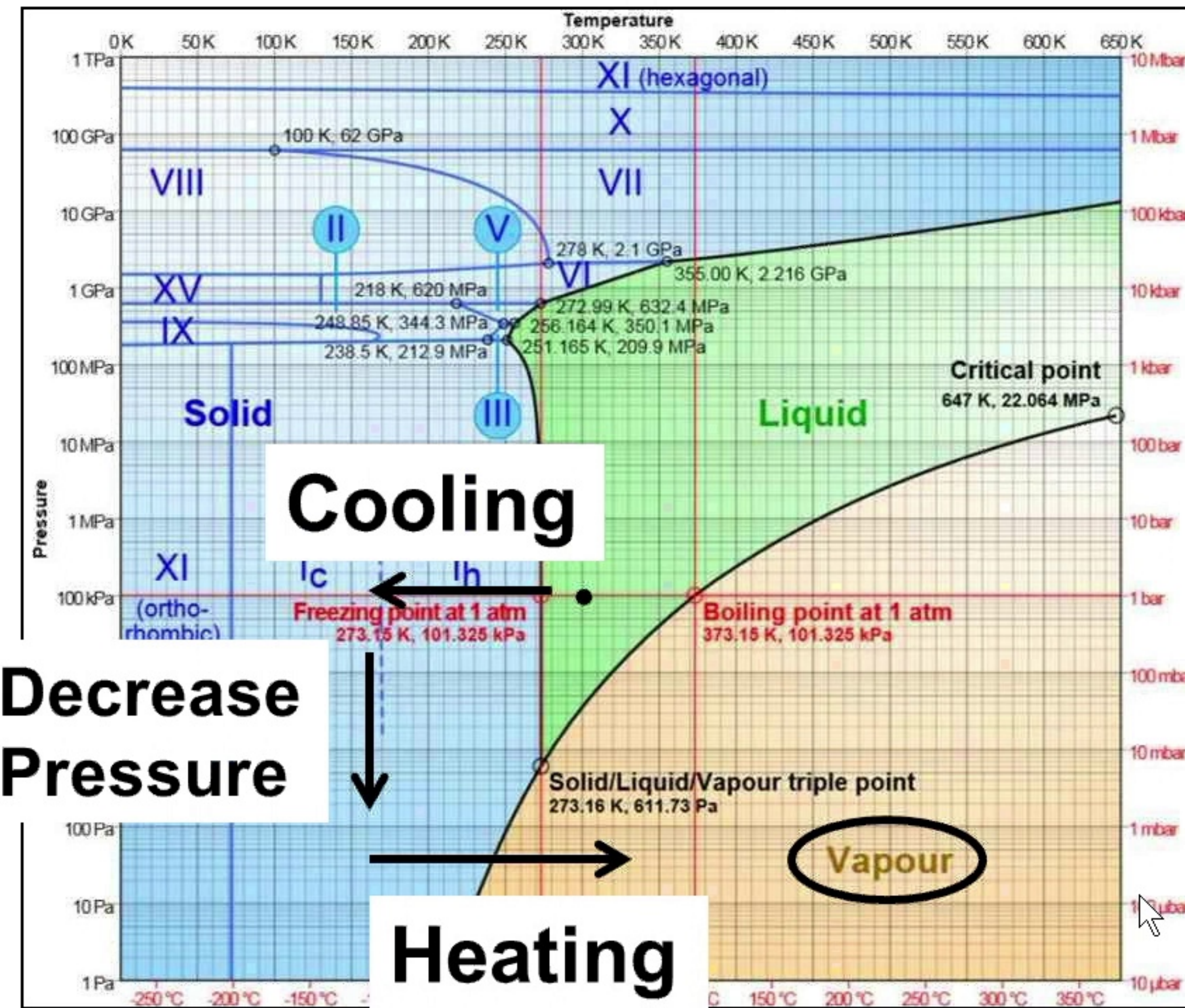
Looking at a Phase Diagram



Goal – Liquid to Vapor: Evaporation



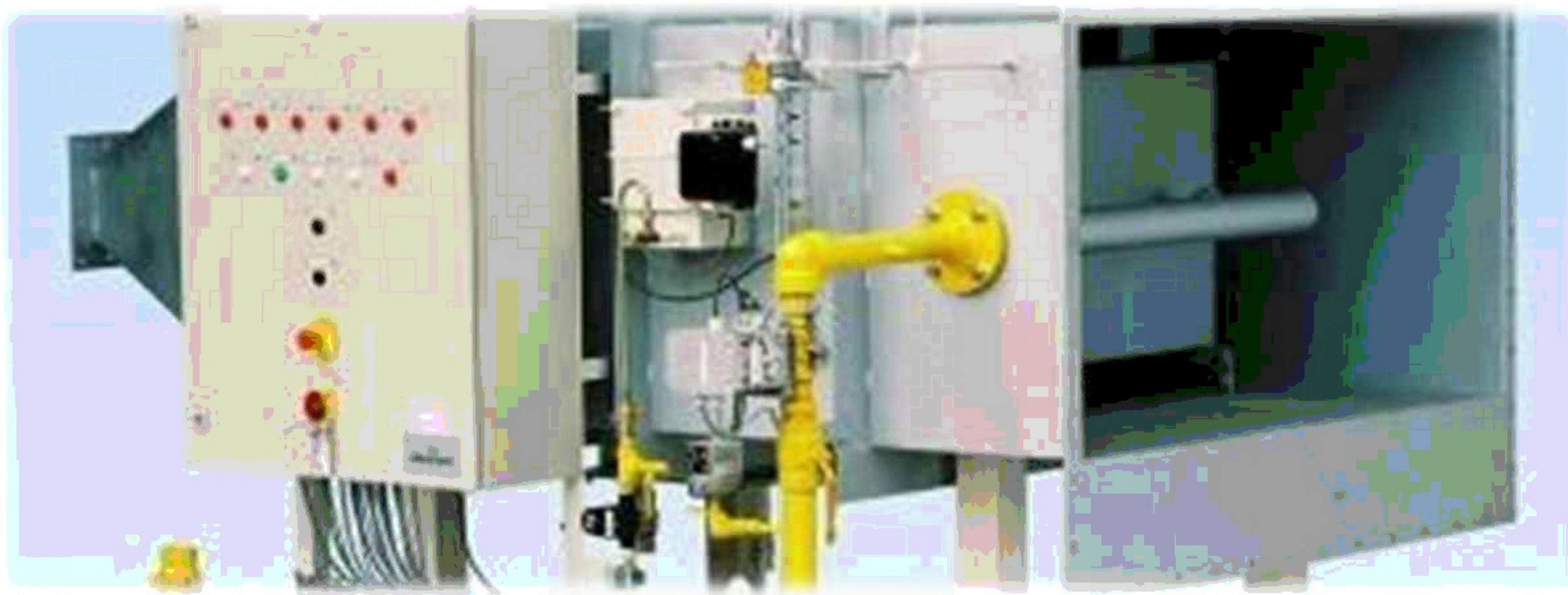
Liquid to Solid to Vapor: Sublimation



Heating and Cooling the System

- As you add energy into the system the temperature increases dependent on the Specific Heat Capacity.
- As you remove energy from the system the product will decrease based the on the Specific Heat Capacity.

$$Energy = \Delta T * C_p * mass$$



Specific Heat Capacities

- **Water**
 - @ 20°C = 4.183 kJ/kg*K
 - @ 50°C = 4.182 kJ/kg*K
 - @ 100°C = 4.219 kJ/kg*K
- **Ethanol**
 - @ 0°C = 2.3 kJ/kg*K
 - @ 40°C = 2.72 kJ/kg*K



Phase Change – Latent Heat ΔH

- The energy required for the solvent to change from one phase to another.
- Liquid to Solid - ΔH_{fus}
- Liquid to Vapor - ΔH_{vap}
- Solid to Vapor- ΔH_{sub}



$$Energy = \Delta H * mass$$



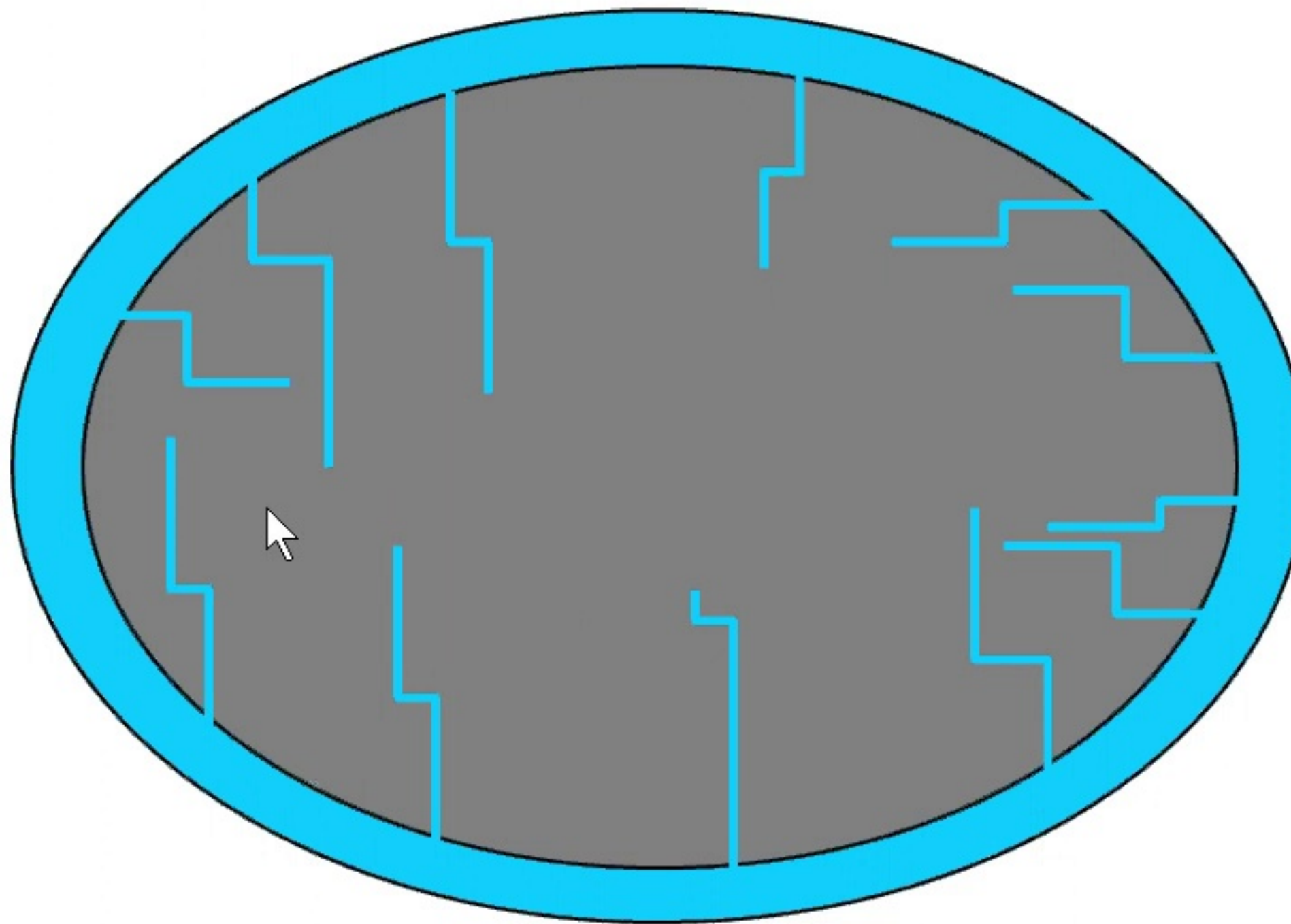
Examples of Latent Heat

- **Ethanol $\Delta H_{\text{vap}} = 846 \text{ kJ/kg}$**
 - Vs. $C_p @ 40^\circ\text{C} = 2.72 \text{ kJ/kg K}$
- **Water $\Delta H_{\text{vap}} = 2257 \text{ kJ/kg}$**
- **Water $\Delta H_{\text{fus}} = 334 \text{ kJ/kg}$**
 - Vs. $C_p @ 100^\circ\text{C} = 4.219 \text{ kJ/kg K}$
 - Assuming $C_p @ 100^\circ\text{C}$ taking 1 kg of water from 0 to 100°C uses 421.9kJ



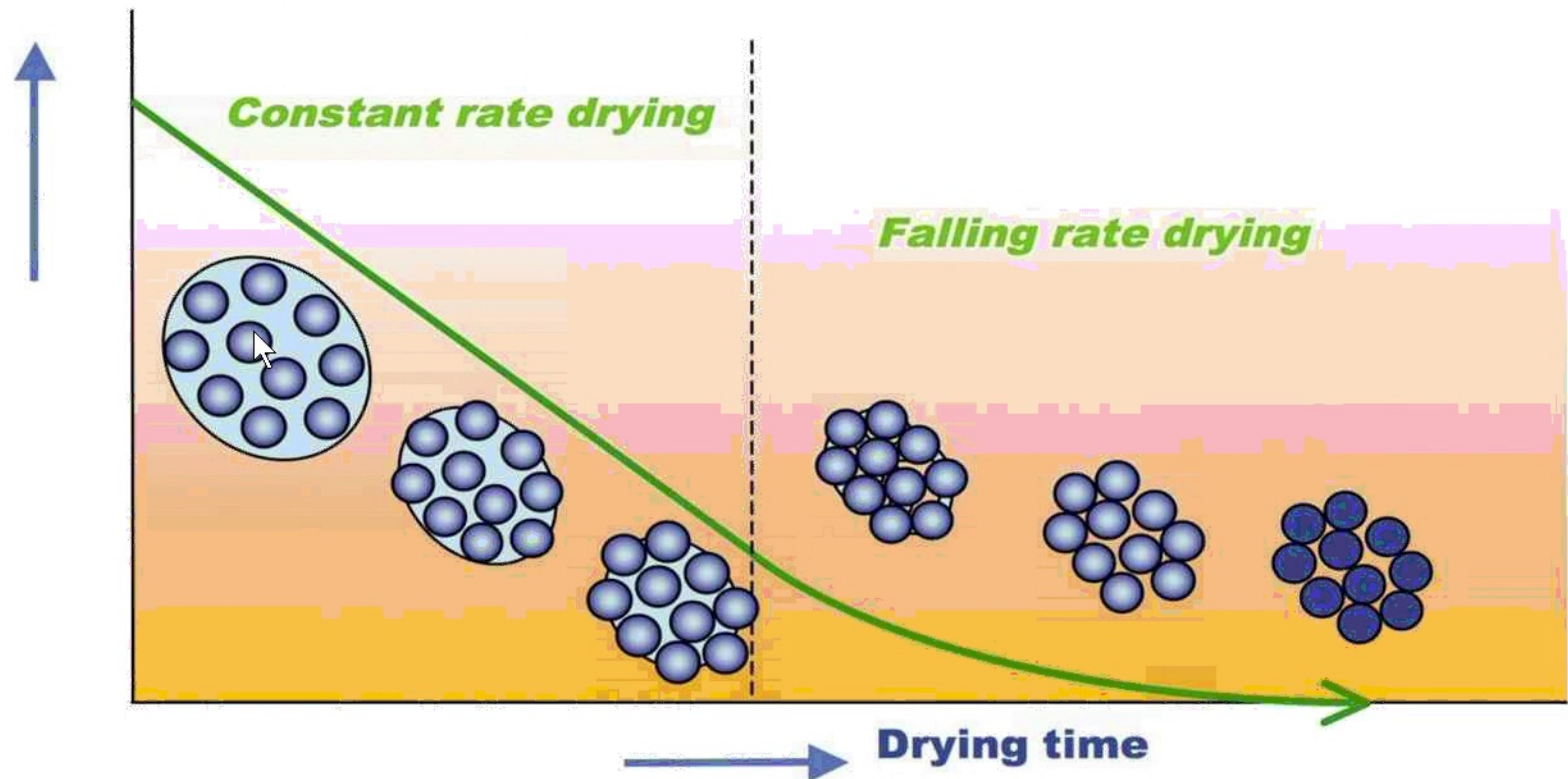
Two types of Moisture

- Surface Moisture
- Bound and Entrapped Moisture



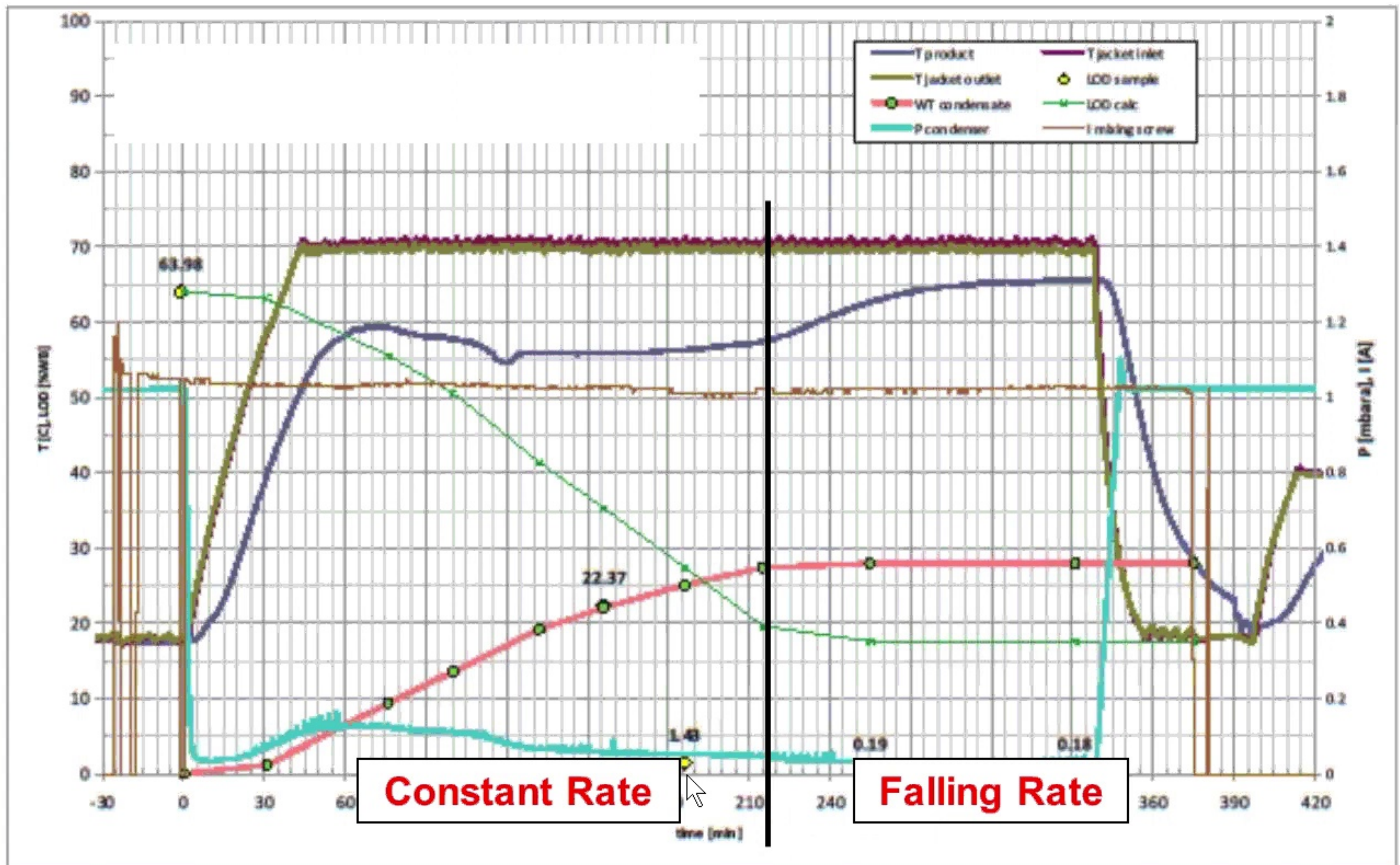
Looking at the Drying Rate

Moisture content
of the product



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Sample Drying Curve



Constant Rate

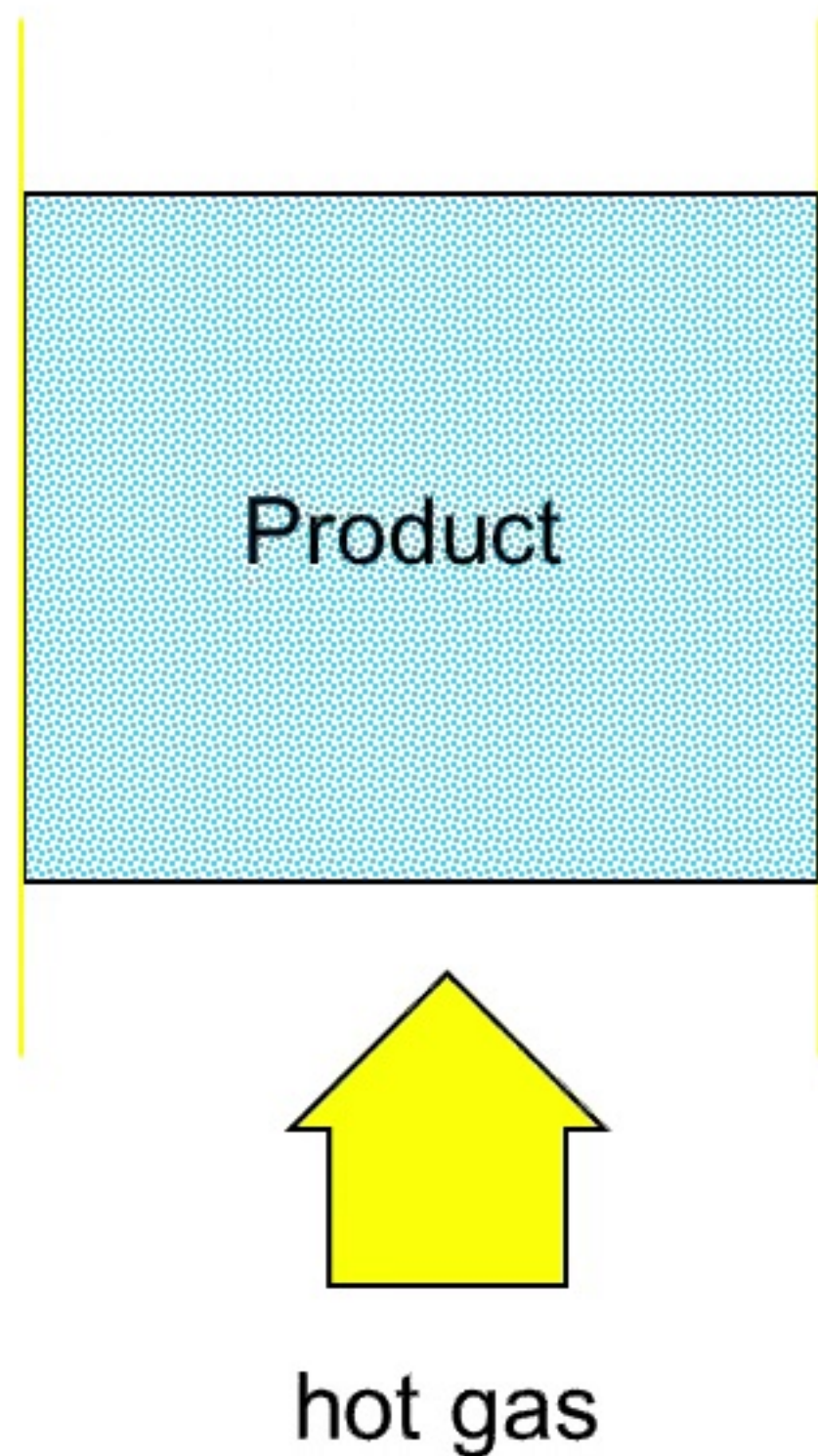
Falling Rate



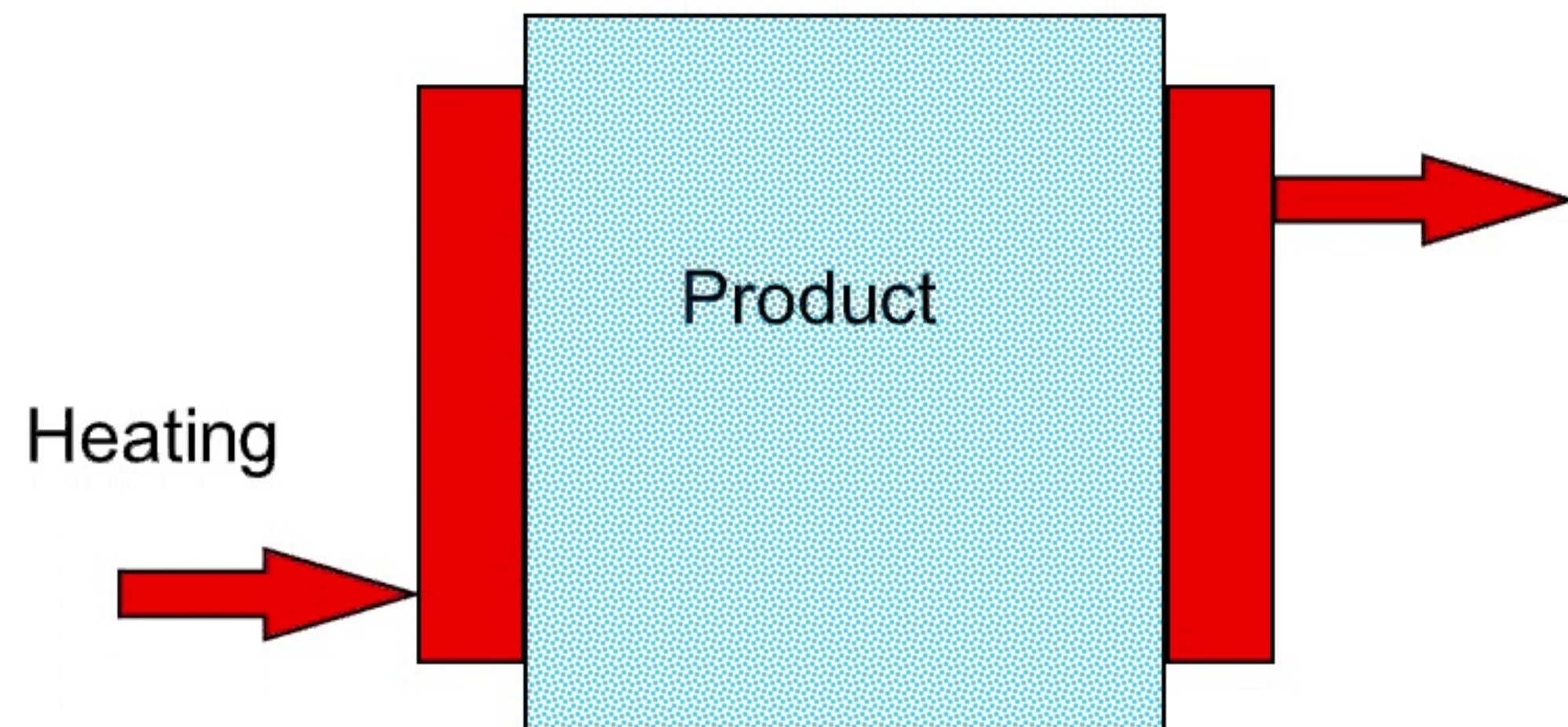
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Direct vs. Indirect Drying

*Direct drying
or
convective drying*



*Indirect drying
or
conductive drying*

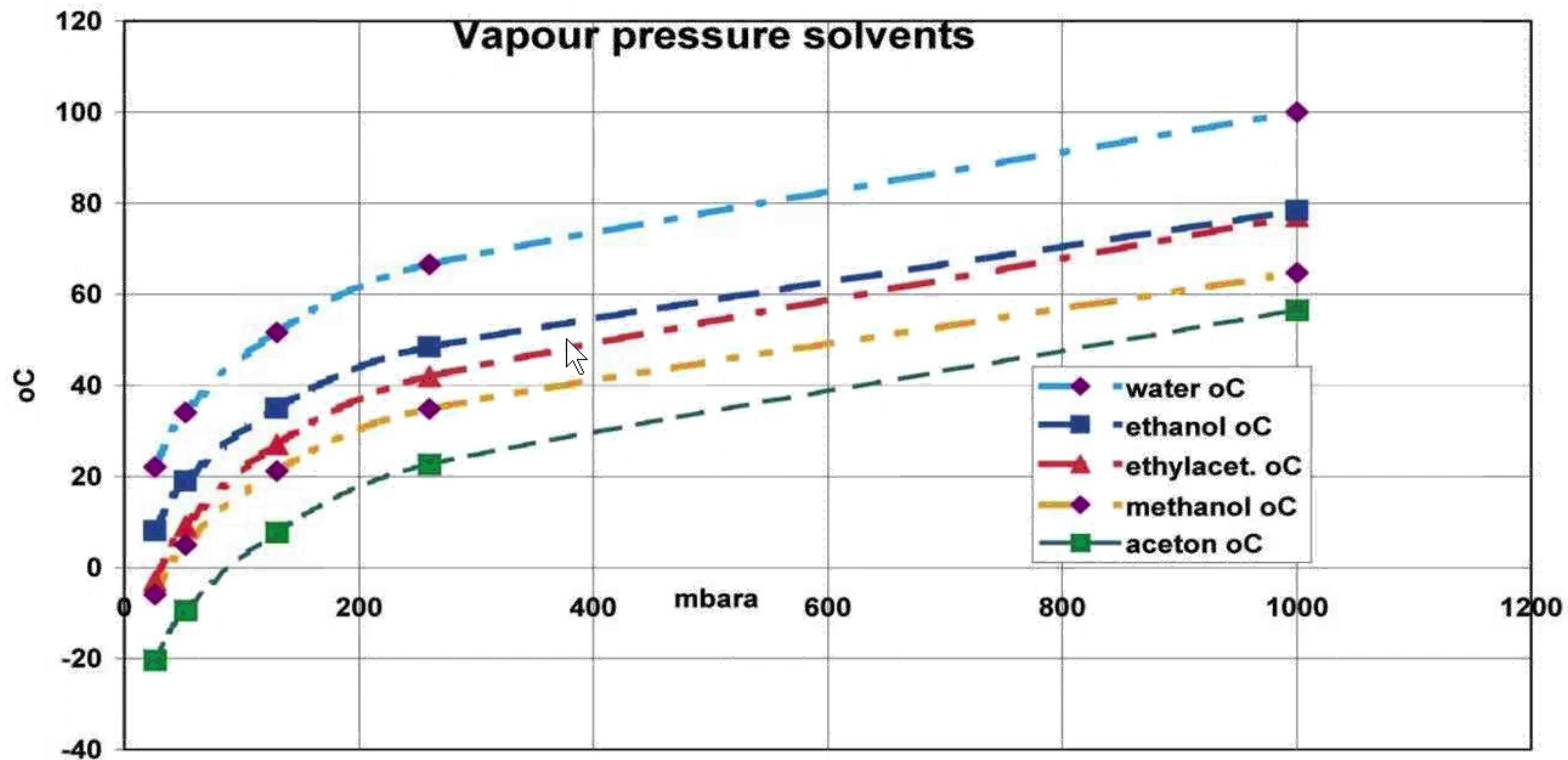


Why Drying at vacuum conditions?

- Drying at low temperature (temperature sensitive products)
- Low end moisture content
- Entrapped liquid
- Controlled and contained drying process
- Solvent recovery
- Toxic, flammable and explosive products



Vacuum Drying Temperature Curves

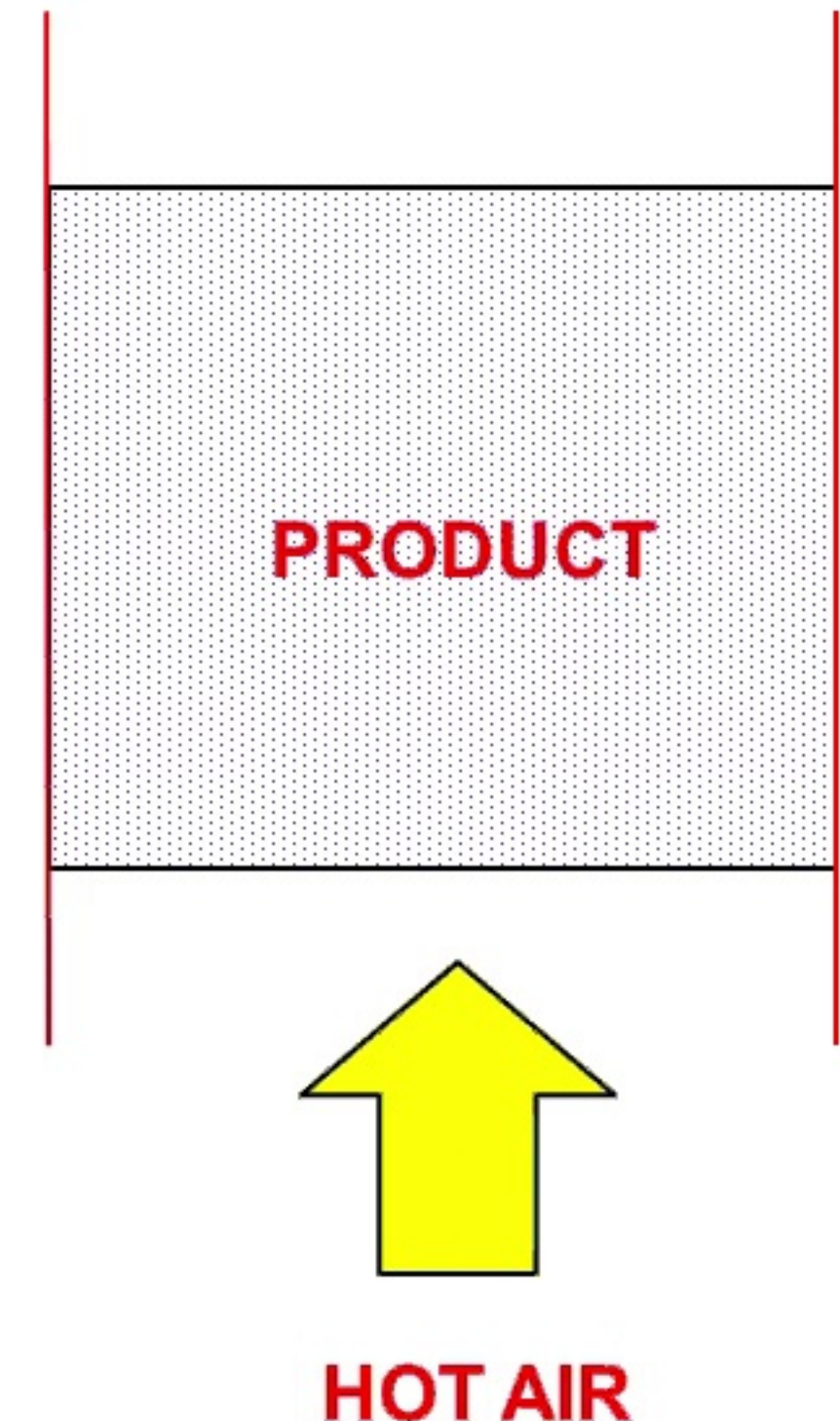


Flash Drying

- **Instantaneous drying by contact with hot airstream, during conveying**

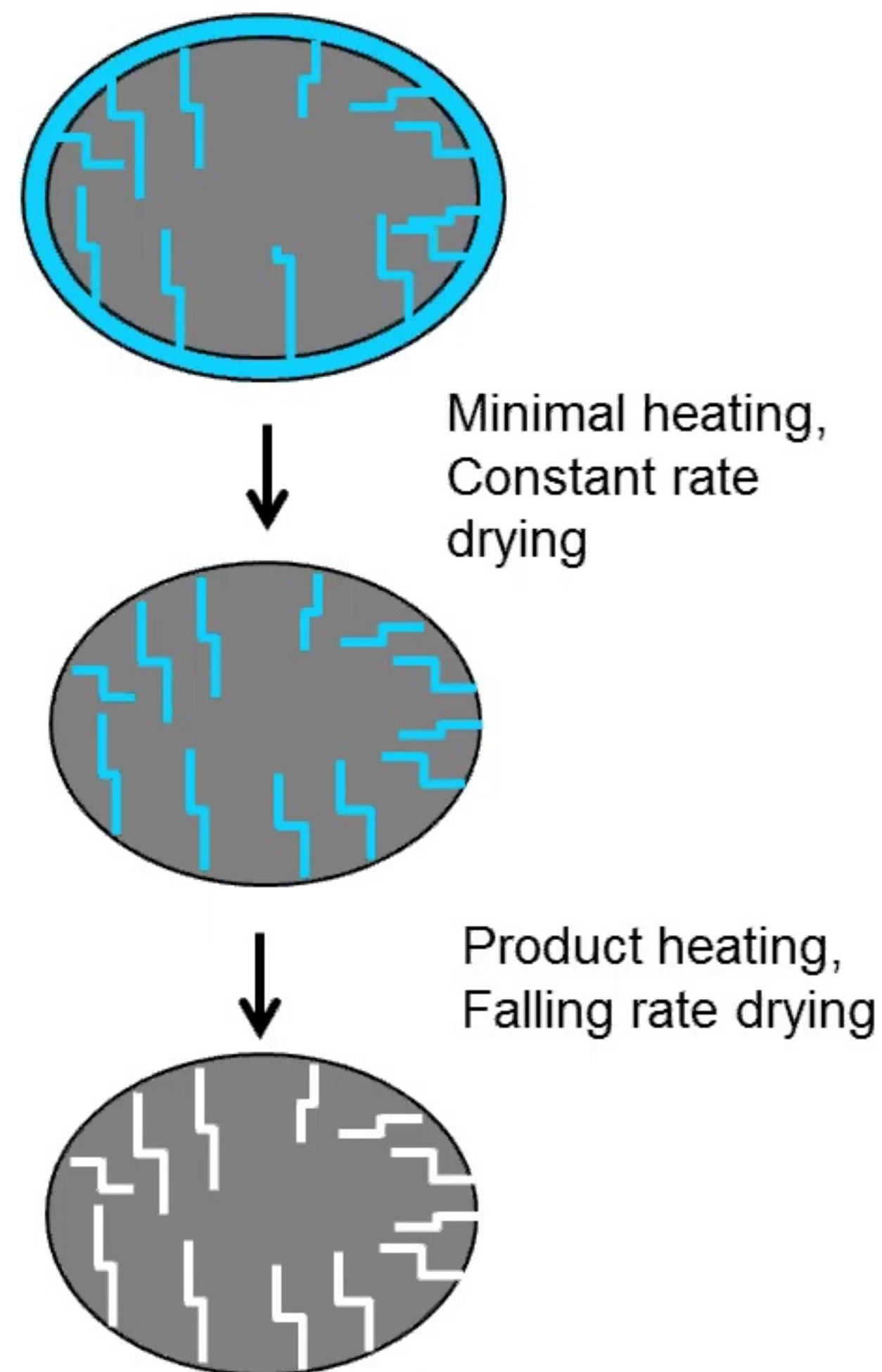
Why Flash drying?

- Large capacities
- High temperatures possible
- End moisture < 5 - < 0.1 %
- Fluidized Bed Dryer
- Spray Tower



Won't my product melt?

- As energy goes from the gas into the system the solid and moisture will heat up
- Once the boiling point of water is reached the system will not heat further until most of the surface moisture is boiled off.
- Obtaining very low final moisture contents for entrapped moisture can require higher temperatures
- 120°C is a good estimate for the outlet air temp and the product should be slightly higher than 100°C
- For low moisture content on products with high entrapped moisture a higher temp out is needed



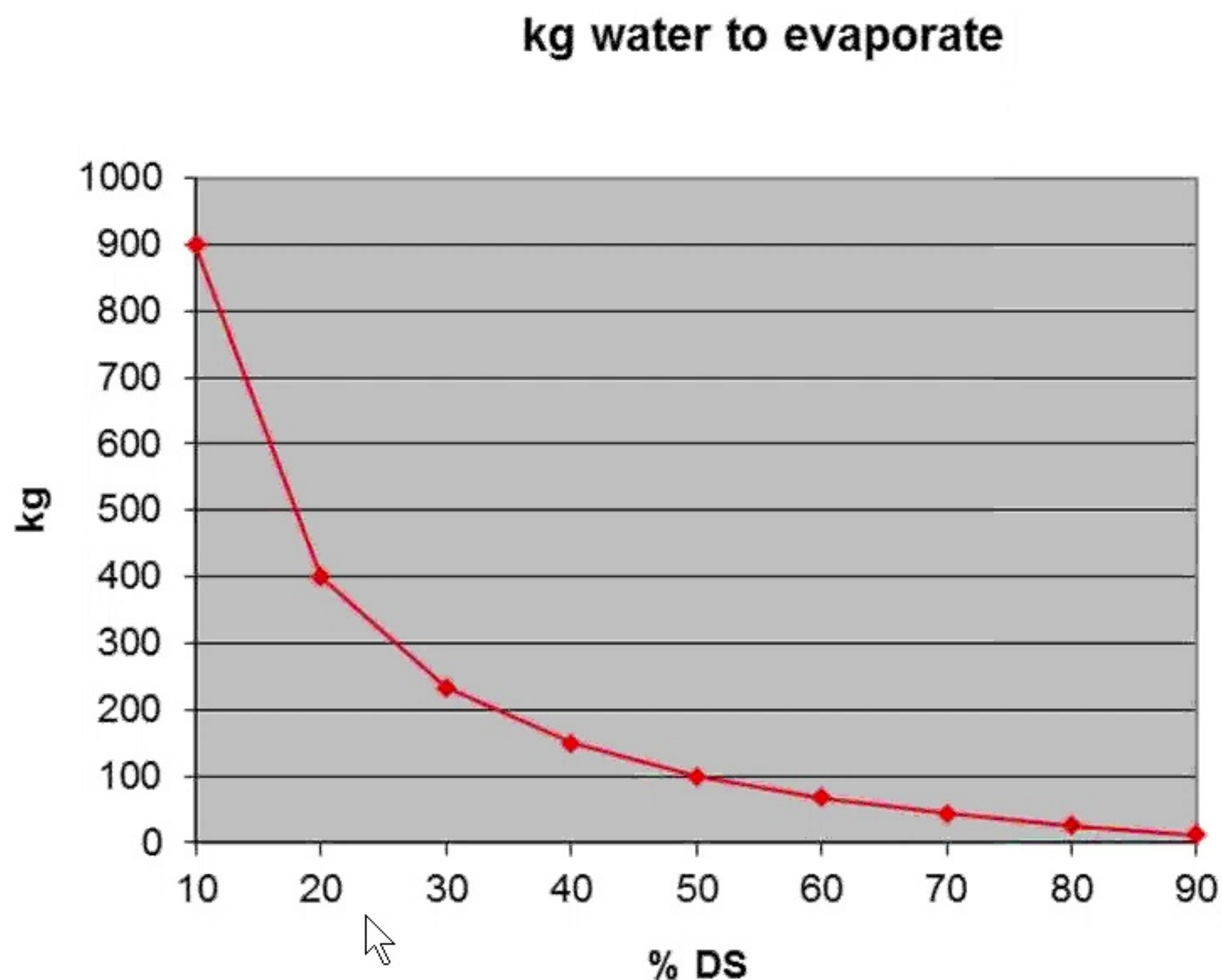
Active Freeze Drying

Why *freeze* drying?

- Ultra-fine (nano-) materials
- Temperature sensitive products
- Delicate structures
- Color or taste
- Living organisms (bacteria, insects)
- Composite materials
- Solutions / Dispersions



Solids Concentration as a variable



kg of water to be evaporated to obtain **100 kg dry material** depending on the feed consistency



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Batch vs. Continuous

- **Continuous**
 - **Flash Drying**
 - **Large(r) throughputs**
 - **Low(er) residence time**
- **Batch**
 - **Vacuum Drying**
 - **Small(er) throughputs**
 - **Long(er) residence times**



Selecting a dryer

Goals and Requirements

Product Based

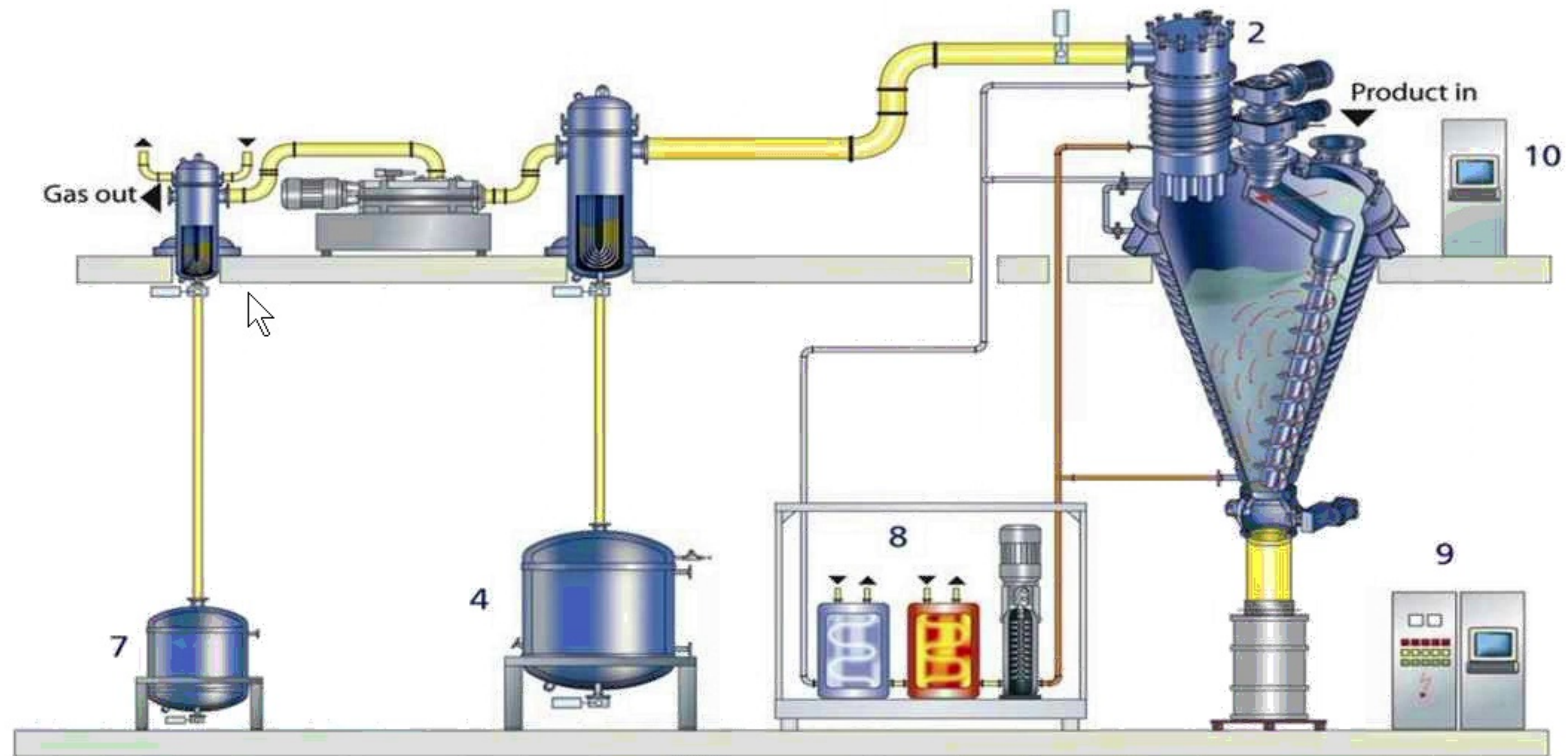
- Product form and feed form
- Solid and liquid properties (f.e. temperatures)
- Initial- and final moisture contents
- Toxicological data
- Safety hazards

Process Based

- Continuous- or batchwise operation
- Throughput and production capacities
- Production changes
- Evaporative load
- Up- and downstream processes and fluctuations



Typical Vacuum Drying System



1. Vacuum dryer
2. Vacuum filter
3. Condenser
4. Receptacle

5. Vacuum pump
6. Optional condenser
7. Optional receptacle
8. Heating cooling skid

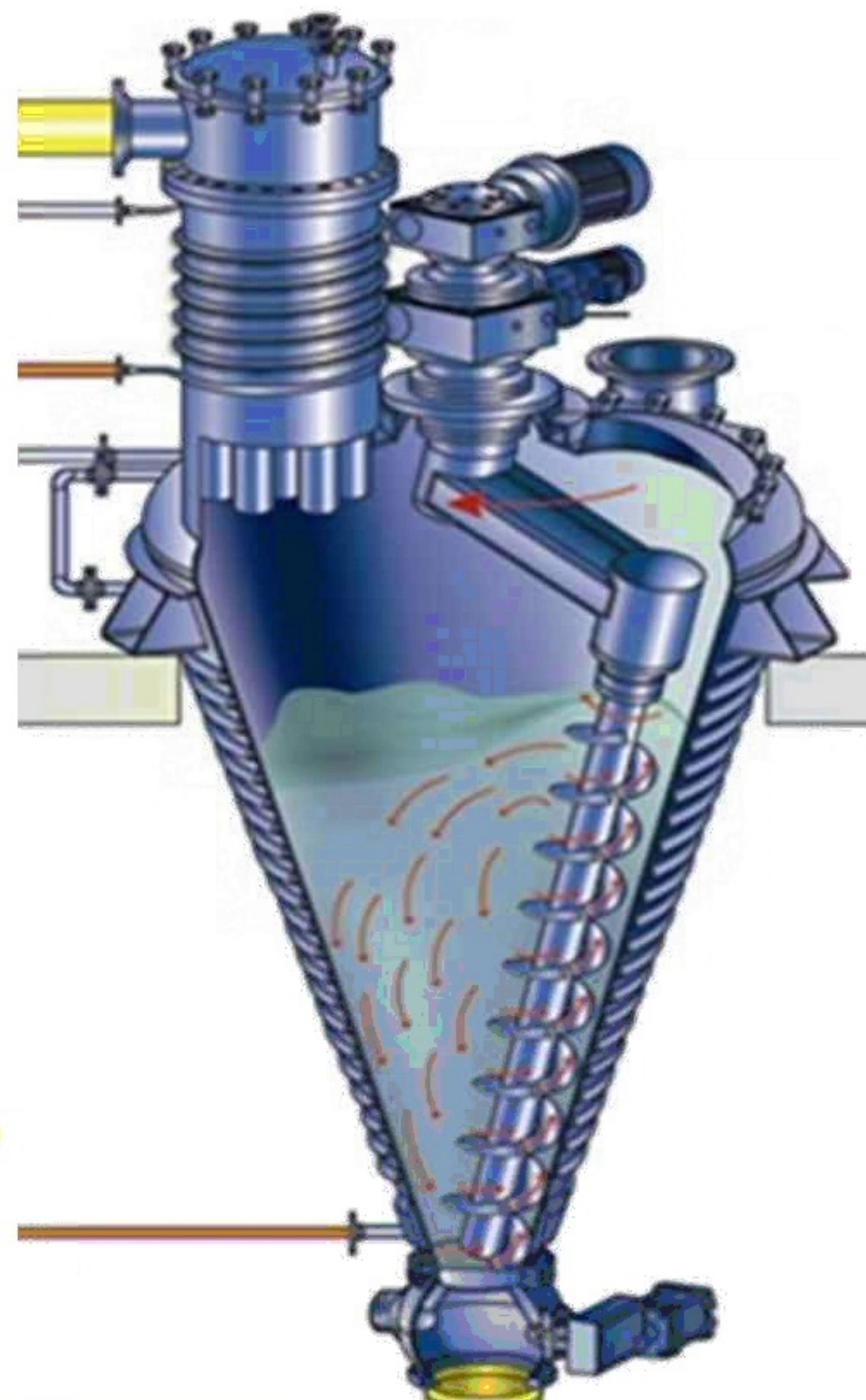
9. Control cabinet
10. Operating panel



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Typical Batch Dryer

- Full Batch in motion means good heat transfer
- Gentle mixing means minimal product breakage
- Handles variable feed stocks (slurries, pastes, and damp powders)



Typical steps in catalyst production

- Preparation of base/carrier material (mixing/drying/homogenisation)
- Impregnation of carrier material with active material (mixing/coating/drying)
- Activating/passivation of active sites (mixing/reacting/thermal treatment)
- (regeneration)



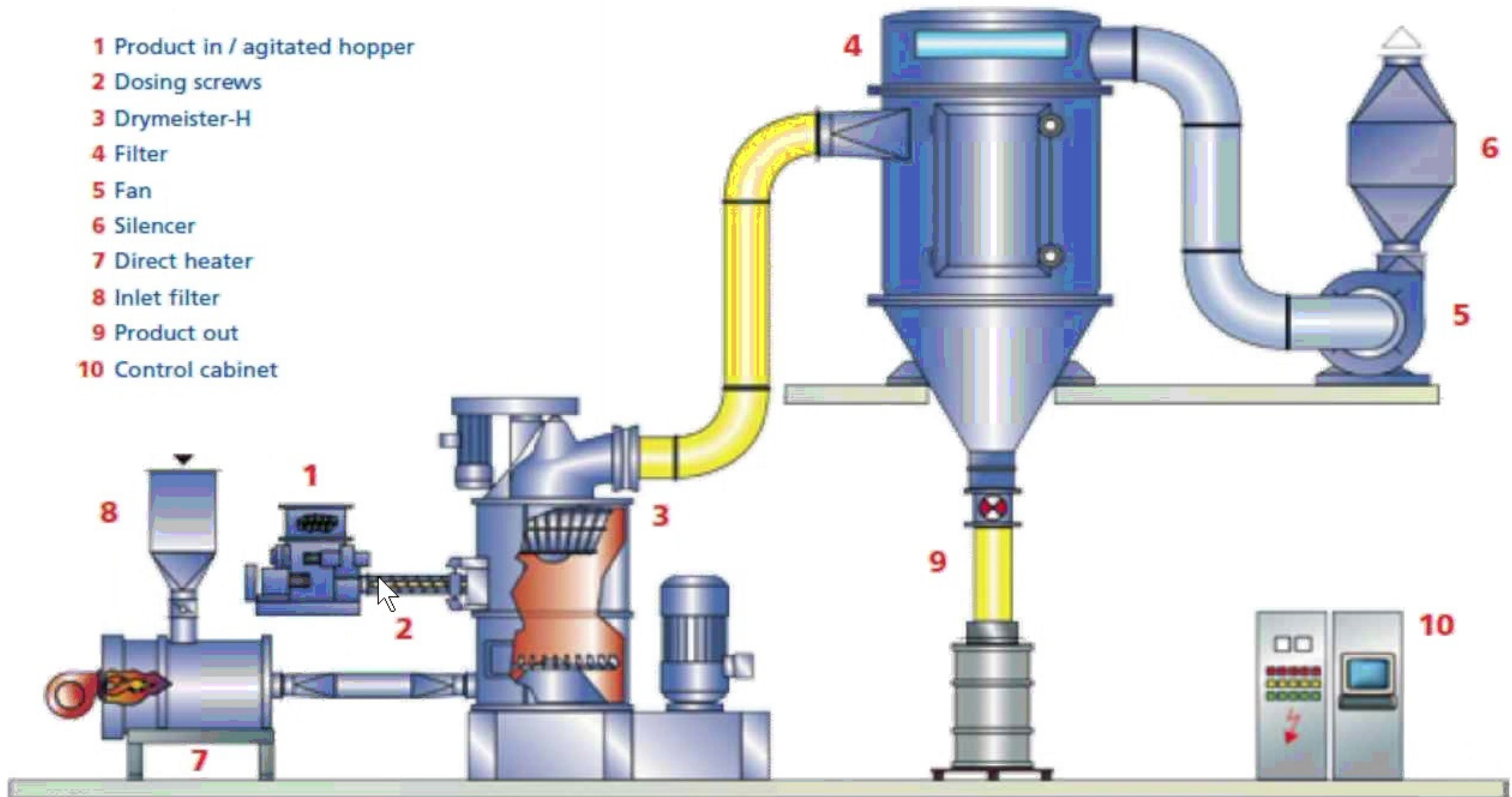
Examples of Applications vacuum drying

- Crop protecting agents
- Catalysts
- Chemically produced toners
- Polymers
- Metal salts
- Nuclear waste
- API's (with advantages of CIP and SIP)
-



Typical Flash Drying System

- 1 Product in / agitated hopper
- 2 Dosing screws
- 3 Drymeister-H
- 4 Filter
- 5 Fan
- 6 Silencer
- 7 Direct heater
- 8 Inlet filter
- 9 Product out
- 10 Control cabinet

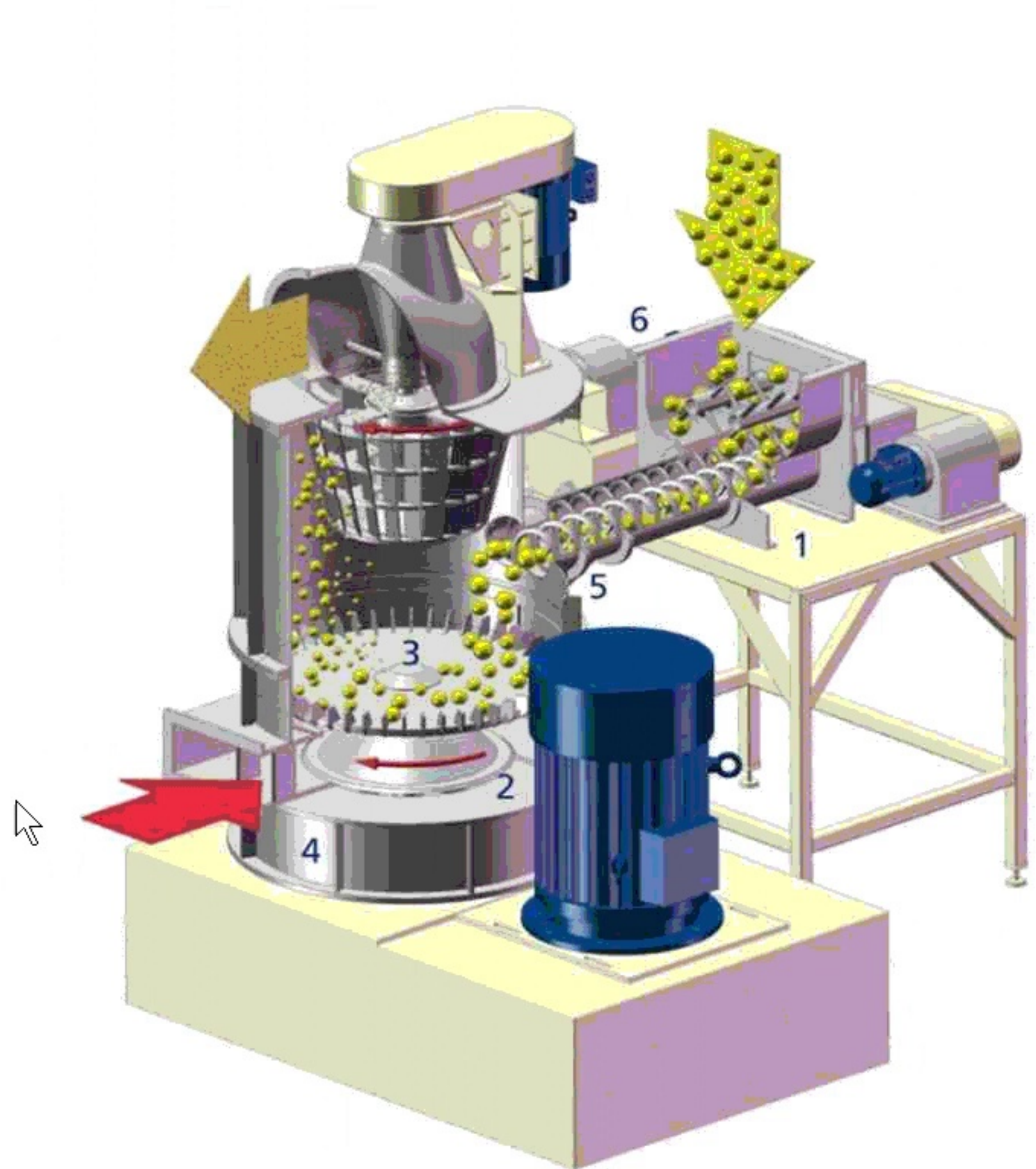
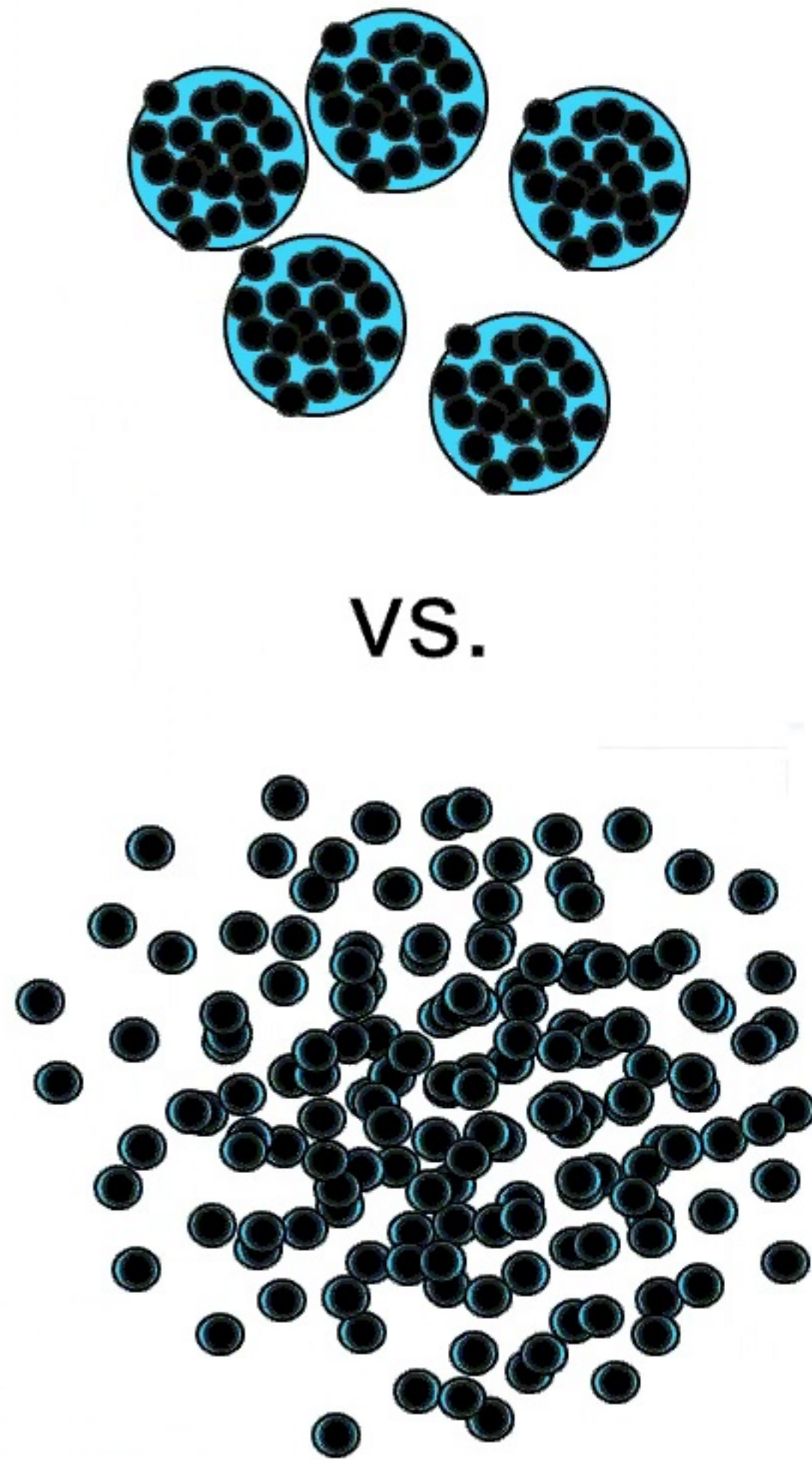


Flow sheet of an open flash drying system



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Typical Flash Dryer



Energy saving

Increased inlet temperature

Sensible heat available per m³ at different inlet temperatures (outlet temperature 120°C)

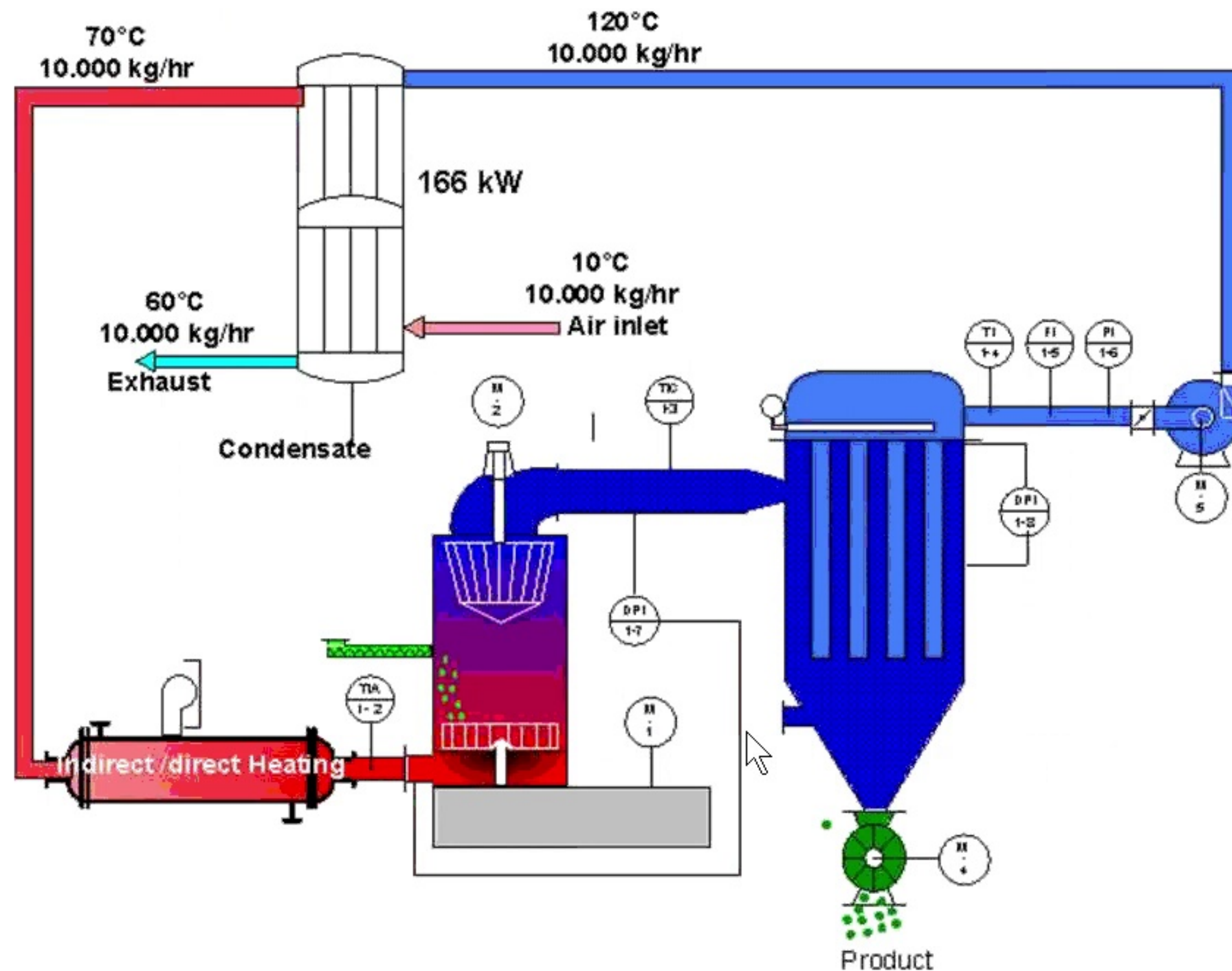


	150°C	250°C	350°C	450°C	550°C	650°C
kJ/m ³	30	130	230	330	430	530



Energy savings

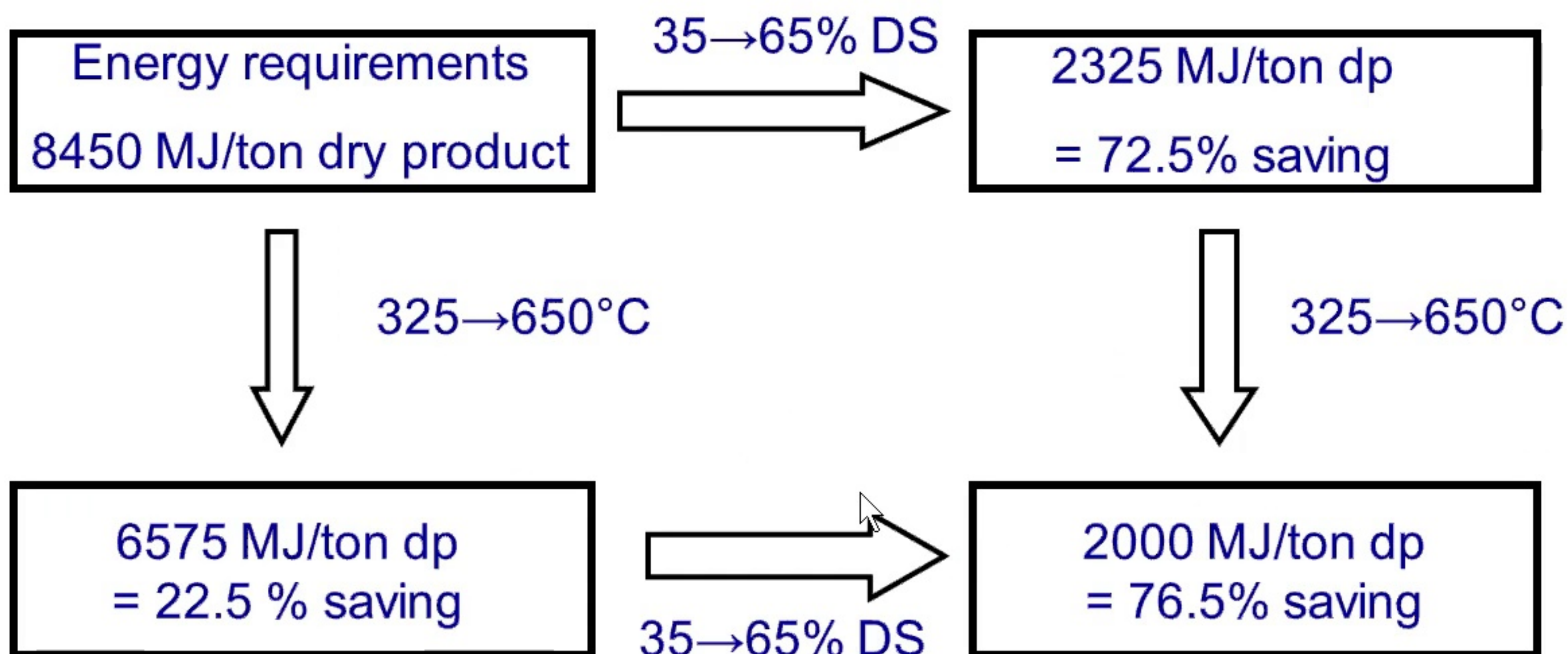
- Reclaim heat from exhaust air using Heat Exchanger



Energy Savings through Process changes

Basis: CaCO_3 with feed 35% DS / out 99.9% DS

Dryer inlet temperature 325°C / outlet 120°C



Applications flash drying

Chemical

- Chromium-yellow
- Coprous-oxide
- Organic pigment
- Inorganic pigment
- Ultramarine Blue
- Biosulphur waste
- Zinc phosphate
- Zeolite
- PCC/GCC
- Magnesium(hydr)oxide

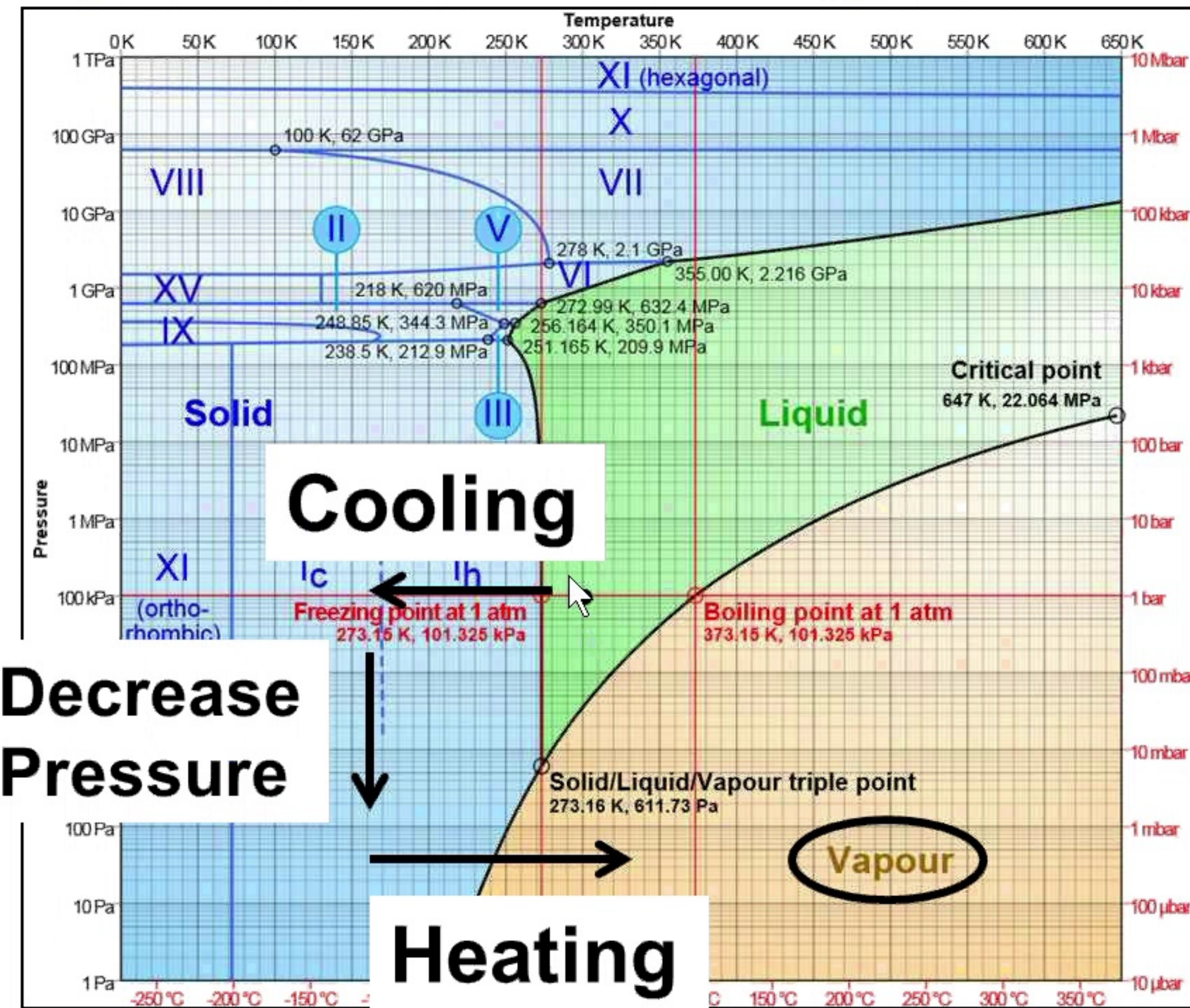
Food

- Okara
- Alginate (Acid)
- Alginate (sodium)
- Flour for bread
- Aspartame
- Natural Colorant
- Yeast
- Flavors
- Fish Meat



Freeze Drying

Goal - Liquid to Solid to Vapor



The art of Freeze Drying



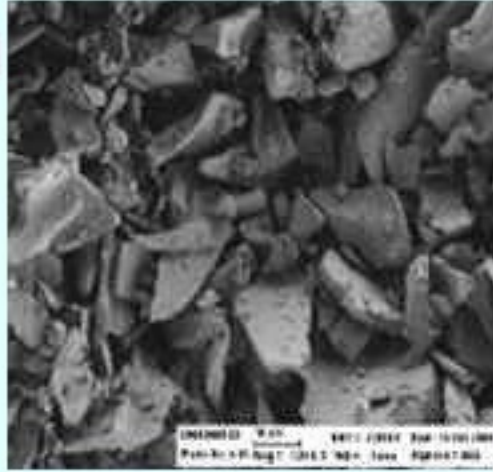
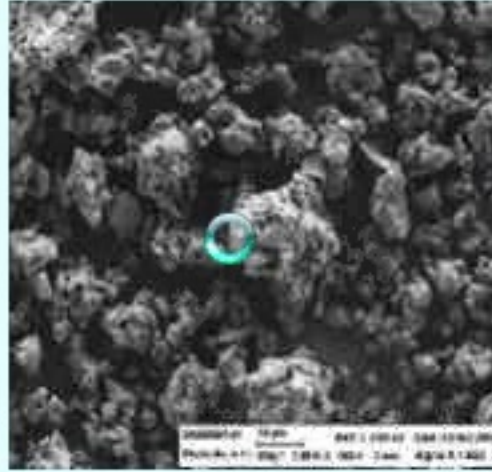
Good: porous structure

Bad: solid structure



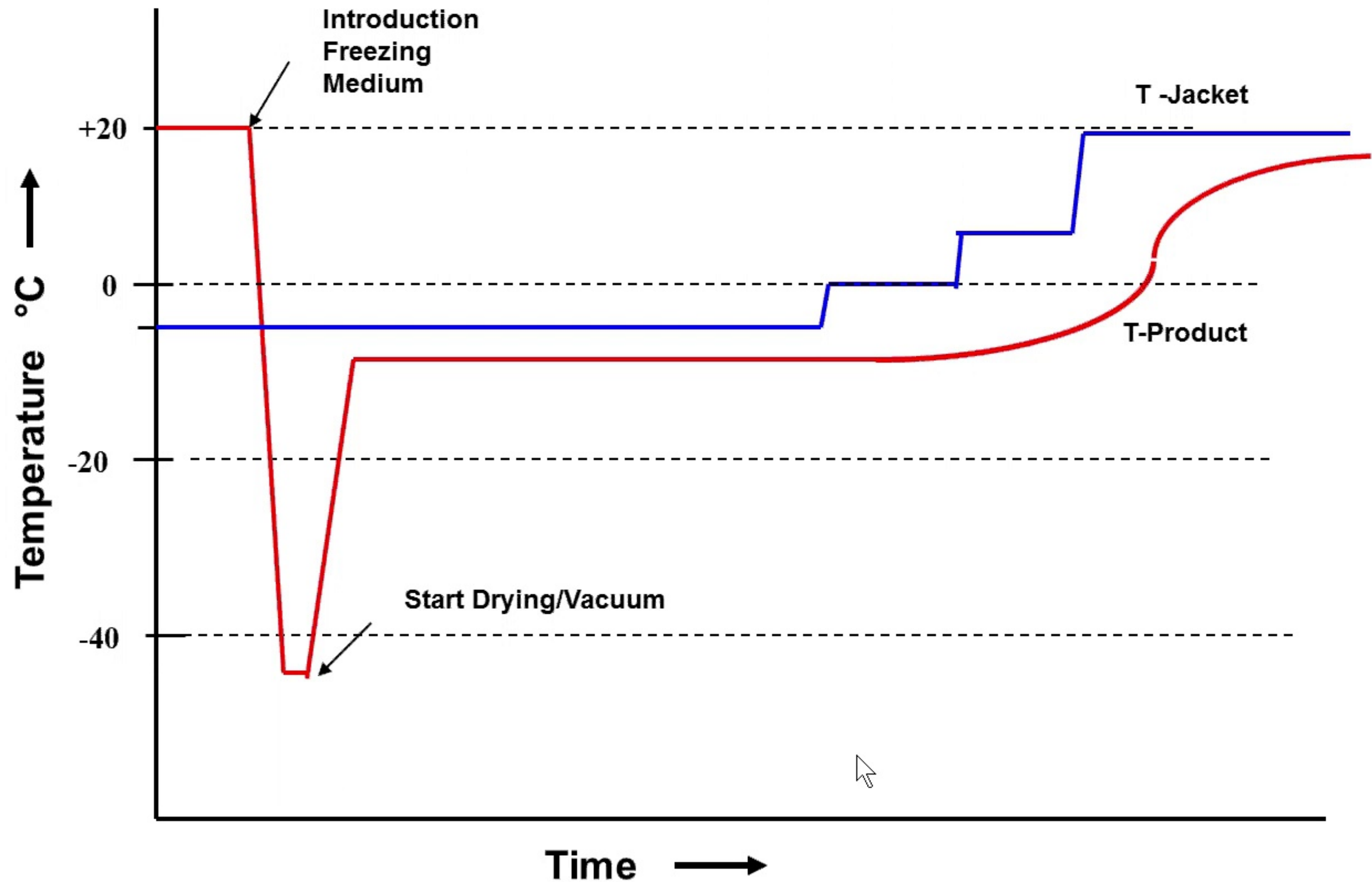
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Conventional Tray drying vs. Active freeze drying

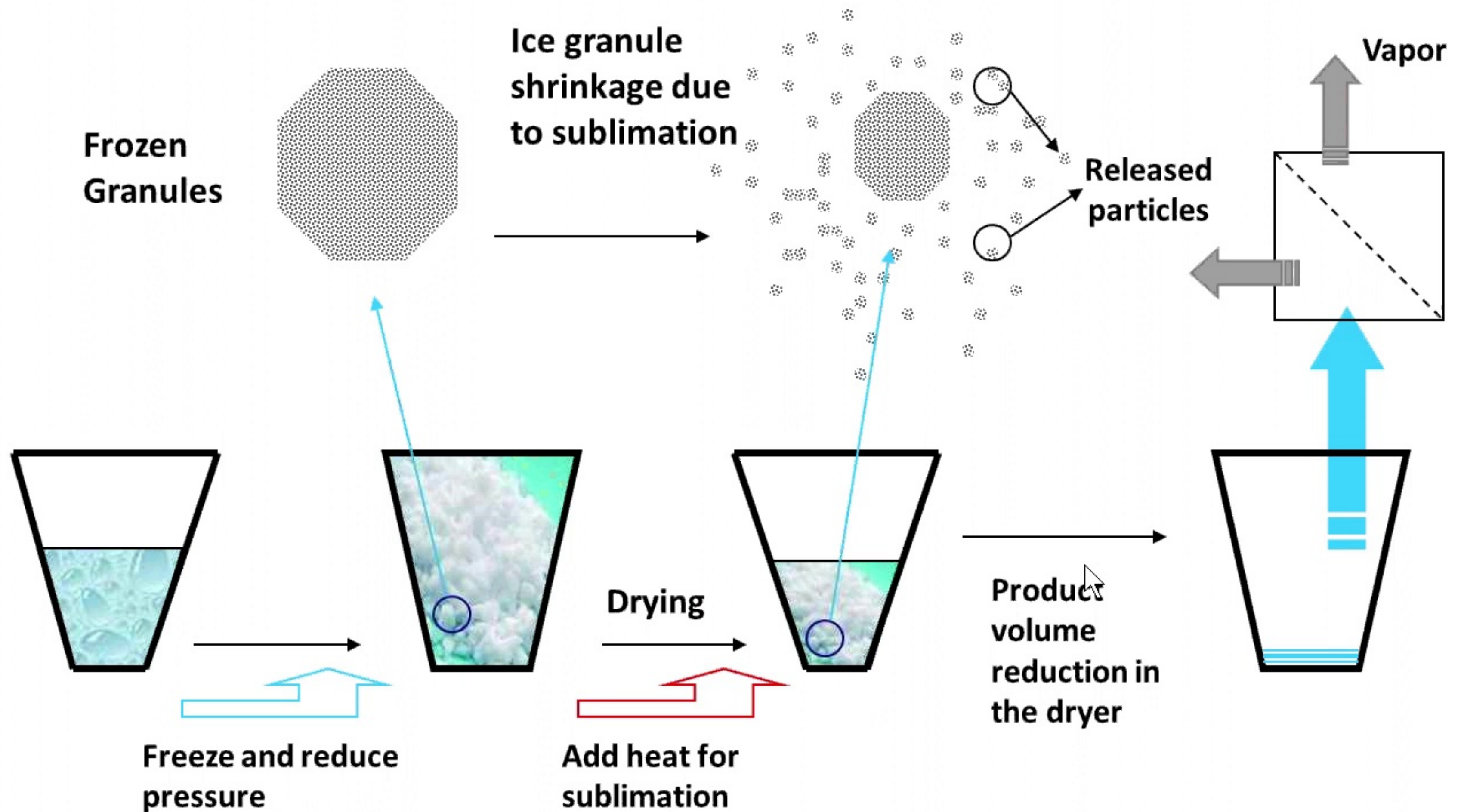
Type of dryer	Conventional tray dryer	Active freeze dryer
Photo's		
Handling	Intensive	Easy
Product in dryer	Product stationary	Product in motion
Shape end product	Big massive lumps (coming out of trays) 	 Powdery
Post treatment	Manual discharge and crush or mill	Automatic discharge from vessel or filter
API (10 μ m)	End product after milling 	End product coming out of freeze dryer 



Temperature-curve Active Freeze Drying



Active Freeze Drying process (general)



Common Freeze Drying Applications

Pharma:

- PLGA
- Lipides
- API's
- Algenates
- Cellulose fibres



Food:

- Enzymes
- Bakery Yeast
- Mushrooms
- Watercress-extract
- Carrots/pumpkins
- Herbal extract
- Camel Milk
- Colostrum
- Corn cream soup
- Flavors
- Instects
- Proteins
- Garlic
- Ginger
- Algae
- Specialty foods



Non-food:

- Nano particles
- Titanium dioxide
- PTFE-dispersion
- Aluminium oxide
- Dental material
- Barium Sulphate
- Glass powder
- Etc.



Compare Technologies

- **Vacuum Drying**
 - Batch operation for precision environment control
 - Gentle operation
 - Low energy requirements
 - CIP & SIP for food and pharmaceutical applications
- **Flash Drying**
 - Continuous drying operation & large capacity
 - Rapid moisture evaporation
 - Low end moisture content
- **Active Freeze Drying**
 - Produces uniform & consistent quality material
 - Well suited for fragile & delicate structures
 - Does not alter material characteristics



Questions?

Chris Paulsworth

Applications Engineer

Hosokawa Micron Powder Systems

cpaulsworth@hmps.hosokawa.com

1(908)277-9272



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